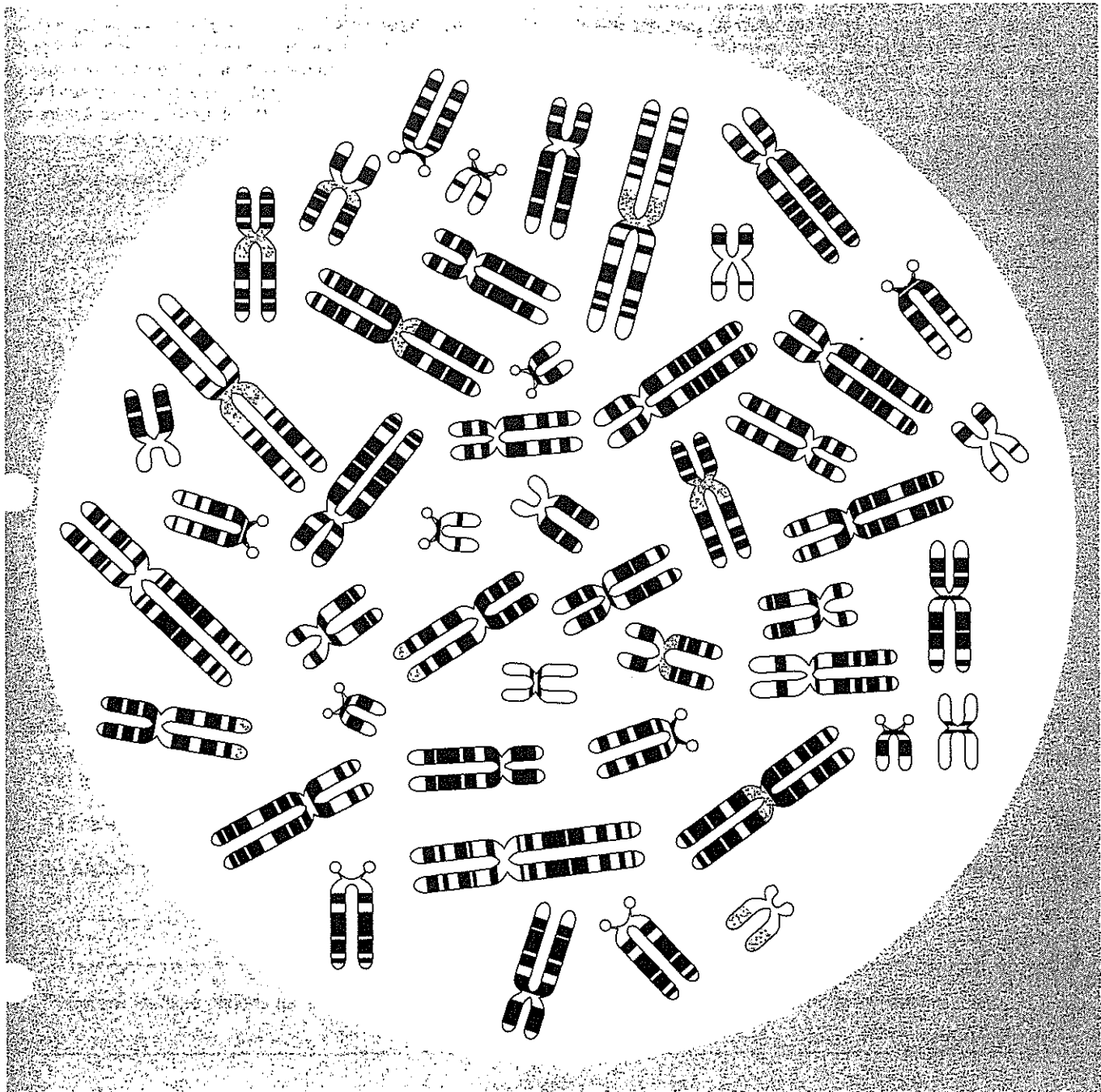
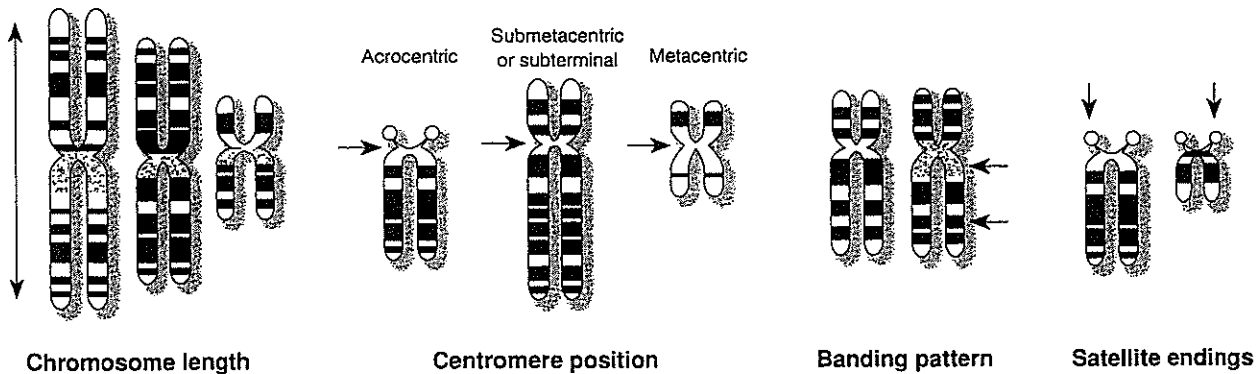


Cut out the chromosomes below and arrange them on the **Record Sheet** provided on the facing page



Distinguishing Characteristics of Chromosomes



Each chromosome has distinctive features that enable it to be identified and distinguished from others. The banding pattern does not represent individual genes, but regions of the chromosome that would contain up to many hundreds of genes. They are 'stained' in a special technique that gives them their banded appearance.



Human Karyotype Exercise

Determine the sex and chromosome condition of the individual whose chromosomes are displayed on the facing page. The karyotypes presented on the previous pages, and

the hints on how to recognise chromosome pairs, can be used to help you complete this activity.

1. Cut out the chromosomes on the facing page and arrange them on this record sheet in their homologous pairs.
2. (a) Determine the sex of this individual: **Male** or **Female** (circle one)
- (b) State whether the individual's *chromosome arrangement* is: **Normal** or **Abnormal** (circle one)
- (c) If the arrangement is *abnormal*, state in what way: _____

1		2		3		4		5	
6		7		8		9		10	
11		12		13		14		15	
16		17		18		19		20	
21		22		Sex chromosomes					

PA 2

Human Karyotype Exercise

Determine the sex and chromosome condition of the individual whose chromosomes are displayed on the facing page. The karyotypes presented on the previous pages, and

the hints on how to recognise chromosome pairs, can be used to help you complete this activity.

1. Cut out the chromosomes on the facing page and arrange them on this record sheet in their homologous pairs.

2. (a) Determine the sex of this individual: **Male** or **Female** (circle one)

(b) State whether the individual's *chromosome arrangement* is: **Normal** or **Abnormal** (circle one)

(c) If the arrangement is *abnormal*, state in what way: There is an extra chromosome 4

